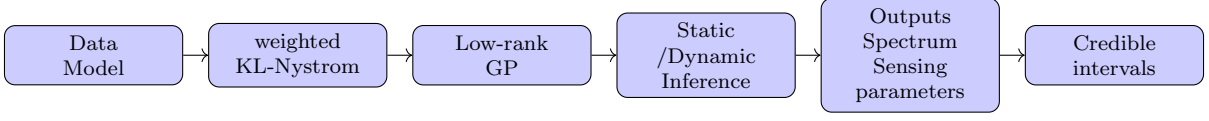


Estimation of Wideband PSD Parameters



PSD Estimation via KL + GP

Data model. Let $\{f_i\}_{i=1}^M$ be a frequency grid with quadrature weights $\{w_i\}_{i=1}^M$ (w.r.t. the Lebesgue measure unless otherwise specified). For wideband spectrum sensing, the default choice is a uniform grid of weights since FFT-based measurements naturally produce uniform frequency grids with spacing Δf , that is $w_i = \Delta f$, $\forall i$. This avoids scaling errors in high-dimensional problems and ensures $\sum_{i=1}^M w_i = M \Delta f = B_{\text{total}}$. Trapezoidal weights, i.e., $w_i = (f_{i+1} - f_{i-1})/2$ with boundary adjustments, are preferred when high accuracy is needed for narrow-band features or non-uniform grids.

Over a given frequency grid, assume $\mathbf{y}$ be a linearized/averaged surrogate of PSD data $|X(f)|^2$, then the observed subsampled measurements are modelled as:

$$\mathbf{y} = \boldsymbol{\Theta} \mathbf{s} + \boldsymbol{\varepsilon}, \quad \mathbf{y} \in \mathbb{R}^N, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon}), \quad (1)$$

where $\mathbf{s} \in \mathbb{R}^M$ is the (unknown) PSD over the frequency grid, $\boldsymbol{\Theta} \in \mathbb{R}^{N \times M}$ is a known linear sampling/sensing operator ($N \leq M$ – subsampling), and $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \boldsymbol{\Sigma}_{\varepsilon})$ is the noise term, having covariance, $\boldsymbol{\Sigma}_{\varepsilon} \in \mathbb{R}^{N \times N}$. A common special case is homoscedastic noise with $\boldsymbol{\Sigma}_{\varepsilon} = \sigma^2 \mathbf{I}_N$.

KL basis construction. Choose a positive definite kernel $k_f: \mathcal{F} \times \mathcal{F} \rightarrow \mathbb{R}$ on frequency. Form the weighted Gram matrix with entries:

$$\mathbf{K}_w[ij] = k_f(f_i, f_j) \sqrt{w_i w_j}, \quad i, j = 1, \dots, M. \quad (2)$$

Compute its eigendecomposition (retain $R \ll M$ dominant modes):

$$\mathbf{K}_w \mathbf{v}_n = \lambda_n \mathbf{v}_n, \quad \lambda_1 \geq \dots \geq \lambda_R > 0, \quad n = 1, \dots, R \quad (3)$$

Define discrete eigenfunctions on the grid

$$\phi_n(f_i) \approx \mathbf{v}_n[i] / \sqrt{w_i}, \quad i = 1, \dots, M. \quad (4)$$

For off-grid evaluation $f \in \mathcal{F}$ (Nyström extension):

$$\phi_n(f) \approx \frac{1}{\lambda_n} \sum_{j=1}^M k_f(f, f_j) w_j \phi_n(f_j). \quad (5)$$

KL feature matrix and low-rank field model. Define the KL feature matrix $\boldsymbol{\Phi}_{\text{KL}} \in \mathbb{R}^{M \times R}$ by its entries

$$(\boldsymbol{\Phi}_{\text{KL}})_{in} := \sqrt{\lambda_n} \phi_n(f_i), \quad i = 1, \dots, M, \quad n = 1, \dots, R. \quad (6)$$

Approximate the PSD on the grid by the rank- R KL expansion

$$\mathbf{s} \approx \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\xi}, \quad \boldsymbol{\xi} \in \mathbb{R}^R, \quad (7)$$

with prior $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_R)$. This induces the (approximate) truncated Mercer/KL covariance on $\mathbf{s}$:

$$\text{Cov}(\mathbf{s}) \approx \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\Phi}_{\text{KL}}^{\top} = \sum_{n=1}^R \lambda_n \boldsymbol{\phi}_n \boldsymbol{\phi}_n^{\top}, \quad (8)$$

where $\boldsymbol{\phi}_n := [\phi_n(f_1), \dots, \phi_n(f_M)]^{\top} \in \mathbb{R}^M$. The measurement model becomes

$$\mathbf{y} \approx \boldsymbol{\Theta} \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\xi} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon}). \quad (9)$$

[Single-window] Static Bayesian linear regression. Let $\mathbf{A} := \boldsymbol{\Theta} \boldsymbol{\Phi}_{\text{KL}} \in \mathbb{R}^{N \times R}$ so that $\mathbf{y} = \mathbf{A} \boldsymbol{\xi} + \boldsymbol{\varepsilon}$. With $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_R)$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon)$, the posterior is

$$\boldsymbol{\Sigma}_{\xi|y} = \left(\mathbf{I}_R + \mathbf{A}^\top \boldsymbol{\Sigma}_\varepsilon^{-1} \mathbf{A} \right)^{-1}, \quad (10a)$$

$$\boldsymbol{\mu}_{\xi|y} = \boldsymbol{\Sigma}_{\xi|y} \mathbf{A}^\top \boldsymbol{\Sigma}_\varepsilon^{-1} \mathbf{y}. \quad (10b)$$

Within a single window, the PSD posterior (on the grid) is Gaussian with

$$\boldsymbol{\mu}_{s|y} = \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\mu}_{\xi|y}, \quad (11a)$$

$$\boldsymbol{\Sigma}_{s|y} = \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\Sigma}_{\xi|y} \boldsymbol{\Phi}_{\text{KL}}^\top. \quad (11b)$$

[Multiple-windows] Dynamic GP prior on coefficients. For windows $t=1, \dots, T$, let $\mathbf{y}_t = \boldsymbol{\Theta}_t \mathbf{s}_t + \boldsymbol{\varepsilon}_t$ and assume $\mathbf{s}_t \approx \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\xi}_t$ with $\boldsymbol{\xi}_t \in \mathbb{R}^R$.

Place independent GP priors across time on coefficients:

$$\xi_n(\cdot) \sim \mathcal{GP}(0, k_n(\cdot, \cdot)), \quad n = 1, \dots, R, \quad (12)$$

e.g., Matérn kernels ($\nu \in \{1/2, 3/2, 5/2\}$).

Use state-space realizations for scalable inference. Particularly, for $\nu=1/2$ (Ornstein–Uhlenbeck state-space form), the discrete-time model for step Δt_t is

$$\xi_{n,t} = a_{n,t} \xi_{n,t-1} + \eta_{n,t}, \quad a_{n,t} = e^{-\alpha_n \Delta t_t}, \quad (13a)$$

$$\eta_{n,t} \sim \mathcal{N}\left(0, q_n \frac{1 - e^{-2\alpha_n \Delta t_t}}{2\alpha_n}\right), \quad (13b)$$

The stationary initial prior is $\xi_{n,1} \sim \mathcal{N}(0, q_n/(2\alpha_n))$, with hyperparameters $\alpha_n, q_n > 0$. This state-space representation enables efficient $\mathcal{O}(TR^2)$ inference via Kalman filtering/smoothing

In the following case of $\nu \in \{3/2, 5/2\}$, use standard p -dimensional Markov state embeddings with known $(\mathbf{F}_n, \mathbf{L}_n, \mathbf{Q}_n)$ (omitted for brevity).

The observation equation is

$$\mathbf{y}_t = \mathbf{H}_t \boldsymbol{\xi}_t + \boldsymbol{\varepsilon}_t, \quad \mathbf{H}_t := \boldsymbol{\Theta}_t \boldsymbol{\Phi}_{\text{KL}}, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}). \quad (14)$$

Run a R -dimensional Kalman filtering with diagonal transition $\mathbf{A} = \text{diag}(a_{n,t})$ and diagonal process noise $\mathbf{Q}_t = \text{diag}(\cdot)$, using Rauch–Tung–Striebel smoothing across t and across modes n , and producing smoothed $\boldsymbol{\mu}_{\xi|y,t}$ and $\boldsymbol{\Sigma}_{\xi|y,t}$, and reconstruct

$$\boldsymbol{\mu}_{s|y,t} = \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\mu}_{\xi|y,t}, \quad \boldsymbol{\Sigma}_{s|y,t} = \boldsymbol{\Phi}_{\text{KL}} \boldsymbol{\Sigma}_{\xi|y,t} \boldsymbol{\Phi}_{\text{KL}}^\top. \quad (15)$$

Mostly, state-space representations reformulate GP regression using Kalman filtering theory, achieving $\mathcal{O}(n)$ complexity for linear-time Markovian processes (plus the cost to handle $\boldsymbol{\Sigma}_{\varepsilon,t}$).

Outputs: Spectral Parameter Estimation

Spectral parameter estimation from the reconstructed PSD. Let $\{f_i\}_{i=1}^M$ be the frequency grid with quadrature weights $\{w_i\}_{i=1}^M$ (for a uniform grid, $w_i = \Delta f$). From the KL+GP inference, assume the (approx.) Gaussian posterior

$$\mathbf{s} \mid \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_{s|y}, \boldsymbol{\Sigma}_{s|y}), \quad \mu_{s|y,i} := \boldsymbol{\mu}_{s|y}[i], \quad \sigma_{s|y,i} := \sqrt{\boldsymbol{\Sigma}_{s|y}[i, i]}.$$

All feature computations assume PSD values are in *linear* power units (not dB); if inputs are in dB, convert via $s^{\text{lin}} = 10^{s^{\text{dB}}/10}$ before applying the definitions below.

PSD credible bands

Band statistic: Let $B \subset \{1, \dots, M\}$ be an index set (band), and define a linear band statistic $p_B := \mathbf{c}_B^\top \mathbf{s}$, $\mathbf{c}_B \in \mathbb{R}^M$. Two common choices for $\mathbf{c}_B$ are:

$$\begin{aligned} \text{(Average PSD over } B) \quad c_B[i] &= \begin{cases} \frac{w_i}{\sum_{j \in B} w_j}, & i \in B, \\ 0, & \text{otherwise,} \end{cases} \\ \text{(Total band power over } B) \quad c_B[i] &= \begin{cases} w_i, & i \in B, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The average PSD choice, p_B , has units of power/Hz; the total power choice, p_B , has units of power. Note that $\sum_{i=1}^M w_i s_i$ approximates $\int s(f) df$.

Pointwise PSD posterior and credible intervals. For a grid index i ,

$$s_i \mid \mathbf{y} \sim \mathcal{N}(\mu_{s|y,i}, \sigma_{s|y,i}^2), \quad \mu_{s|y,i} := \mu_{s|y}[i], \quad \sigma_{s|y,i} := \sqrt{\Sigma_{s|y}[i, i]}.$$

A $(1 - \alpha)$ pointwise credible interval is

$$\mu_{s|y,i} \pm z_{1-\alpha/2} \sigma_{s|y,i}, \quad z_p := \Psi^{-1}(p),$$

where $\Psi(\cdot)$ is the standard normal CDF.

Band statistic posterior and credible intervals. Because p_B is linear in $\mathbf{s}$, under the Gaussian posterior for $\mathbf{s}$ we have

$$p_B \mid \mathbf{y} \sim \mathcal{N}(\mu_B, \sigma_B^2), \quad \mu_B := \mathbf{c}_B^\top \mu_{s|y}, \quad \sigma_B^2 := \mathbf{c}_B^\top \Sigma_{s|y} \mathbf{c}_B.$$

A $(1 - \alpha)$ credible interval for p_B is $\mu_B \pm z_{1-\alpha/2} \sigma_B$.

Occupancy probabilities (per-bin and per-band). Given thresholds τ_i (per-bin) and τ_B (per-band), the posterior occupancy probability is defined as:

$$\begin{aligned} \pi_i &:= \Pr(s_i > \tau_i \mid \mathbf{y}) = 1 - \Psi\left(\frac{\tau_i - \mu_{s|y,i}}{\sigma_{s|y,i}}\right), \\ \pi_B &:= \Pr(p_B > \tau_B \mid \mathbf{y}) = 1 - \Psi\left(\frac{\tau_B - \mu_B}{\sigma_B}\right). \end{aligned}$$

Decision rule: declare bin i (or band B) occupied if the corresponding posterior probability $\pi_i \in [0, 1]$ exceeds a target (e.g., 0.5 or another application-driven threshold).

Threshold calibration under noise-only (example). Let s_{nf} denote the (approximately white) *noise-floor PSD level* (in linear units, e.g., W/Hz) under a noise-only reference; s_{nf} is a property of the background spectrum $\mathbf{s}$ and should not be confused with the *measurement-error* covariance Σ_ε in $\mathbf{y} = \mathbf{\Theta}\mathbf{s} + \varepsilon$. For the average-PSD statistic over B with $|B|$ bins, under a noise-only reference model with (white) noise PSD s_{nf} and an independent-bin normal approximation,

$$\tau_B \approx s_{\text{nf}} \left(1 + \sqrt{\frac{2}{|B|}} Q^{-1}(P_{\text{fa}}) \right), \quad Q(x) := 1 - \Psi(x),$$

which yields $\Pr(\text{false alarm}) \approx P_{\text{fa}}$. A consistent per-bin choice is obtained by setting $|B|=1$:

$$\tau_i \approx s_{\text{nf}} \left(1 + \sqrt{2} Q^{-1}(P_{\text{fa},i}) \right).$$

Calibrate analogously for the total-power statistic if that definition of $\mathbf{c}_B$ is used.

plug-in rule: Replace $\pi_i \geq \pi_0$ by $\mu_{s|y,i} > \tau_i$.

Peak frequency. A simple (single-peak) estimate is

$$i_\star := \arg \max_{1 \leq i \leq M} \mu_{s|y,i}, \quad \hat{f}_{\text{peak}} := f_{i_\star}.$$

If you want the peak *within* occupied bins, use $i_\star := \arg \max_{i \in \hat{\mathcal{O}}} \mu_{s|y,i}$ (when $\hat{\mathcal{O}} \neq \emptyset$).

Spectral centroid (center frequency). The power-weighted centroid (optionally restricted to occupied bins) is

$$\hat{f}_c := \sum_{i \in \hat{\mathcal{O}}} f_i w_i \mu_{s|y,i} / \sum_{i \in \hat{\mathcal{O}}} w_i \mu_{s|y,i}, \quad (\text{use } \hat{\mathcal{O}} = \{1, \dots, M\} \text{ if no occupancy gating}).$$

Occupied bandwidth. If $\hat{\mathcal{O}} \neq \emptyset$, define the occupied support endpoints

$$\hat{f}_{\min} := \min_{i \in \hat{\mathcal{O}}} f_i, \quad \hat{f}_{\max} := \max_{i \in \hat{\mathcal{O}}} f_i, \quad \hat{B}_{\text{occ}} := \hat{f}_{\max} - \hat{f}_{\min}.$$

For multiple disjoint emitters, compute *connected components* of $\hat{\mathcal{O}}$ (contiguous index runs) to obtain $\{\hat{B}_{\text{occ}}^{(k)}\}_k$ and peaks per component.

Uncertainty-aware alternative (expected occupied bandwidth). For a uniform grid ($w_i = \Delta f$), a soft estimate is

$$\mathbb{E}\{B_{\text{occ}} \mid \mathbf{y}\} \approx \sum_{i=1}^M \pi_i \Delta f \quad (\text{or } \sum_i \pi_i w_i \text{ for non-uniform grids}).$$

SNR from reconstructed PSD (noise-floor estimated from unoccupied bins). Let π_N be a small cutoff (e.g., $\pi_N = 0.1$) and define a noise-bin set

$$\hat{\mathcal{N}} := \{i : \pi_i \leq \pi_N\}.$$

Estimate the (approximately white) noise floor (PSD level) by a weighted average

$$\hat{s}_n := \sum_{i \in \hat{\mathcal{N}}} w_i \mu_{s|y,i} / \sum_{i \in \hat{\mathcal{N}}} w_i.$$

Then estimate *signal power above the noise floor* over occupied bins as

$$\hat{P}_{\text{sig}} := \sum_{i \in \hat{\mathcal{O}}} w_i (\mu_{s|y,i} - \hat{s}_n)_+, \quad (x)_+ := \max(x, 0),$$

and the corresponding noise power in the same support as

$$\hat{P}_{\text{noise}} := \hat{s}_n \sum_{i \in \hat{\mathcal{O}}} w_i.$$

Finally, define the SNR (linear and dB):

$$\widehat{\text{SNR}} := \frac{\hat{P}_{\text{sig}}}{\hat{P}_{\text{noise}}}, \quad \widehat{\text{SNR}}_{\text{dB}} := 10 \log_{10}(\widehat{\text{SNR}}).$$

If a calibrated noise-floor PSD level s_{nf} is already available (e.g., from hardware calibration or empty-band measurements), the value $\hat{s}_n := s_{\text{nf}}$ can be set and skip $\hat{\mathcal{N}}$.

(0)Credible intervals for these features. To propagate posterior uncertainty, draw samples $\mathbf{s}^{(m)} \sim \mathcal{N}(\boldsymbol{\mu}_{s|y}, \boldsymbol{\Sigma}_{s|y})$ (or from the KL-latent posterior) and compute

$$\{f_{\text{peak}}^{(m)}, f_c^{(m)}, B_{\text{occ}}^{(m)}, \text{SNR}^{(m)}\}_{m=1}^M$$

Report empirical quantiles (e.g., 2.5% – 97.5%) as credible intervals.